Dimensionality Reduction and PCA-Based Classification Study on MNIST dataset

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1. Introduction to the dataset

MNIST stands for **Modified National Institute of Standards and Technology** dataset. It is a benchmark dataset of handwritten digits (0–9), designed to test pattern recognition and machine learning algorithms.

MNIST contains 70,000 grayscale images. Size of each image is 28 x 28 pixels. Hence there are total 784 features. each feature contains values between 0 to 255 depending on the intensity of the colour. 0 for black and 255 for white, rest transitioned from black to white. Each image is paired with a label from 0 to 9 describing the digits they represent.

$$\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N} \tag{1}$$

where:

- $N = 70{,}000$ is the total number of samples.
- $x_i \in \mathbb{R}^{28\times 28}$ is the grayscale image of the i-th handwritten digit.
- $y_i \in \{0, 1, 2, \ldots, 9\}$ is the corresponding class label.

Each image x_i is stored as a 28x28 matrix.

$$x_i = \begin{bmatrix} p_{1,1} & \cdots & p_{1,28} \\ \vdots & & \\ p_{28,1} & \cdots & p_{28,28} \end{bmatrix}$$

Where each pixel $p_{j,k} \in \{0, 1, 2, \ldots, 255\}$.
Dividing the dataset into the features (X) and labels (y).

$$X \in \mathbb{R}^{N\times 784}, \qquad y \in \mathbb{R}^{N} \tag{2}$$

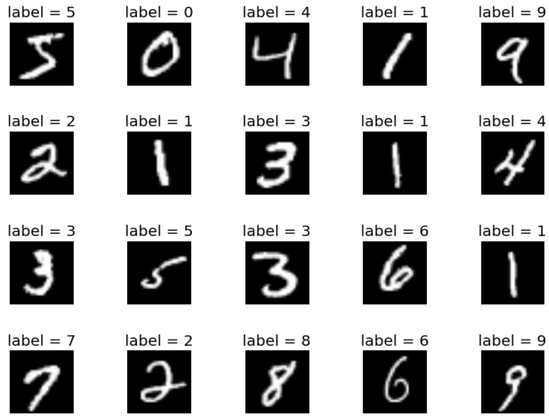

Figure 1: Sample handwritten digit images with corresponding class labels

2. Standardization

Standardization is very important, otherwise the PCA analysis is dominated by the bright pixels(large values) and the background pixels become meaningless noise. PCA will focus on brightness and not on the meaningful inter-pixel correlations.

Standardization ensures that all the features are having unit variance, all the features will contribute equally to the variance analyzed by the PCA.

Standardization is done by the following equation.

$$X_{ij}^{(s)} = \frac{X_{ij} - \mu_j}{\sigma_j} \tag{3}$$

where:

- μ_j is the feature wise mean of the dataset.

$$\mu_j = \frac{1}{N} \sum_{i=1}^{N} X_{ij} \tag{4}$$

- σ_j is the feature wise standard deviation of the dataset.

$$\sigma_j = \sqrt{\frac{1}{N-1} \sum_{i=1}^{N} (X_{ij} - \mu_j)^2} \tag{5}$$

3. Principal Component Analysis (PCA)

3.1 Covariance (Correlation) Matrix

Since the features are standardized, PCA is effectively performed on the covariance matrix:

$$\boldsymbol{C} = \frac{1}{N-1}\boldsymbol{Z}^\top \boldsymbol{Z}, \qquad \boldsymbol{C} \in \mathbb{R}^{784\times 784}. \tag{6}$$

where:

- $\boldsymbol{Z}$ is the standardized data matrix.

3.2 Eigen-Decomposition

PCA is obtained from the eigen-decomposition of $\boldsymbol{C}$:

$$\boldsymbol{C}\boldsymbol{v}_k = \lambda_k \boldsymbol{v}_k, \qquad k = 1, 2, \ldots, 784, \tag{7}$$

where:

- $\lambda_k \geq 0$ are eigenvalues (variances along principal directions),
- $\boldsymbol{v}_k \in \mathbb{R}^{784}$ are orthonormal eigenvectors (principal axes),
- eigenvalues are sorted $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_{784}$.

Let

$$\boldsymbol{V} = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_{784} \end{bmatrix} \in \mathbb{R}^{784\times 784}, \qquad \boldsymbol{\Lambda} = \operatorname{diag}(\lambda_1, \lambda_2, \ldots, \lambda_{784}). \tag{8}$$

Then

$$\boldsymbol{C} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^\top. \tag{9}$$

3.2.1 Principal Component Scores (Low-Dimensional Representation)

Choosing the top K principal components, define

$$\boldsymbol{V}_K = \begin{bmatrix} \boldsymbol{v}_1 & \boldsymbol{v}_2 & \cdots & \boldsymbol{v}_K \end{bmatrix} \in \mathbb{R}^{784\times K}. \tag{10}$$

The PCA embedding (scores) of all samples is

$$\boldsymbol{T} = \boldsymbol{Z}\boldsymbol{V}_K \in \mathbb{R}^{N\times K}. \tag{11}$$

3.3 Explained Variance Ratio

The total variance in standardized space is

$$\operatorname{tr}(\boldsymbol{C}) = \sum_{k=1}^{784} \lambda_k. \tag{12}$$

The explained variance ratio of the k-th component is

$$\mathrm{EVR}_k = \frac{\lambda_k}{\sum_{j=1}^{784} \lambda_j}. \tag{13}$$

3.4 Scree Plot

The scree plot visualizes the eigenvalues λ_k in descending order to illustrate the variance captured by each principal component. A sharp decay followed by a gradual flattening indicates the intrinsic low-dimensional structure of the dataset.

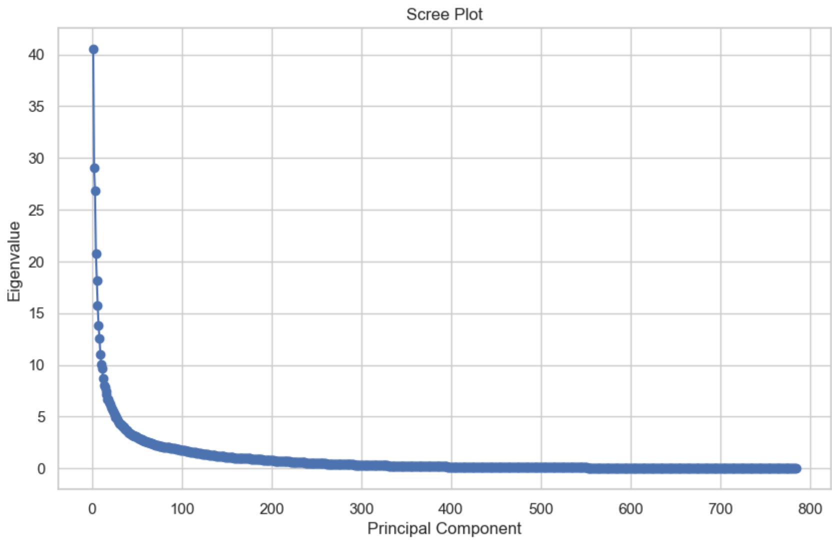

Figure 2: Scree plot showing eigenvalues λ_k versus principal component index.

3.5 Cumulative Explained Variance

The cumulative explained variance curve is defined as:

$$\mathrm{CEV}(K) = \sum_{k=1}^{K} \frac{\lambda_k}{\sum_{j=1}^{784} \lambda_j} \tag{14}$$

This curve helps determine the minimum number of principal components required to retain a desired percentage of total variance (e.g., 95%). The variance distribution is summarized below:

- Principal component 1 (λ_1) accounts for 5.64% of total variance.
- $\sum_{k=1}^{5} \mathrm{EVR}_k = 18.85\%$ i.e first five principal components holds 18.85% of the total variance.
- $K = 149$ components retain 80% variance.
- $K = 237$ components retain 90% variance.

- $K = 330$ components retain 95% variance.

This gradual variance accumulation suggests that handwritten digit images exhibit distributed structural variability rather than dominance by a few principal directions.

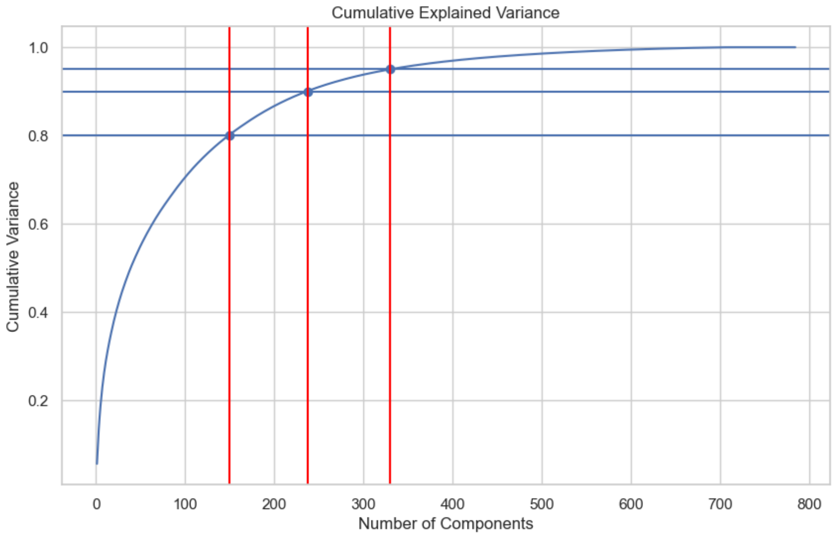

Figure 3: Cumulative explained variance as a function of the number of principal components.

4. Effect of PCA Dimensionality on Classification Accuracy

For each selected number of principal components K, the training data were projected onto the first K principal directions. The classifier was trained on the reduced feature space and evaluated on the test set. The resulting accuracy values are shown in Figure 4.

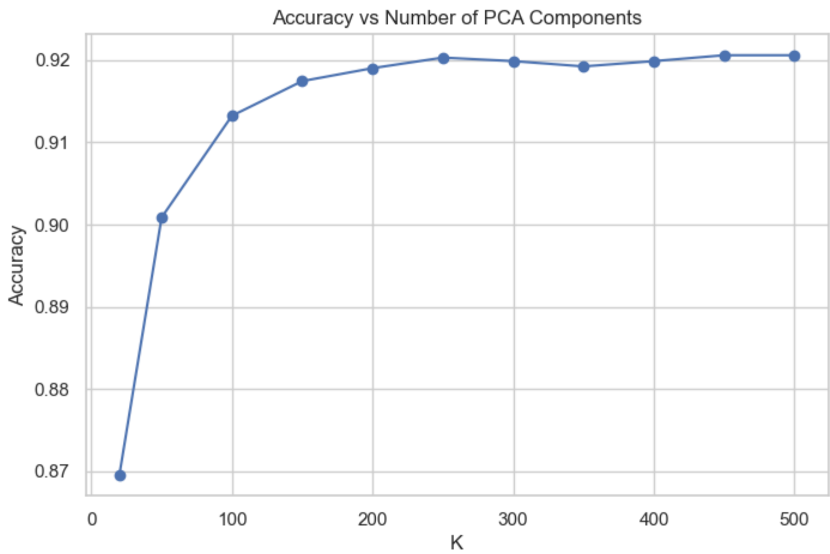

Figure 4: Classification accuracy as a function of the number of principal components K.

The results indicate that classification accuracy increases rapidly for small values of K, followed by a gradual saturation beyond approximately $K \approx 250$. Although retaining 95% of total variance requires 330 components, near-optimal classification performance is achieved with fewer dimensions. This demonstrates that discriminative information is concentrated in a lower-dimensional subspace and does not necessarily needs total variance retention.

5. Reconstruction (Inverse Transform)

Given the PCA scores $\boldsymbol{T}$, the rank-K reconstruction in standardized space is

$$\widehat{\boldsymbol{Z}}_K = \boldsymbol{T}\boldsymbol{V}_K^\top = \boldsymbol{Z}\boldsymbol{V}_K\boldsymbol{V}_K^\top, \qquad \widehat{\boldsymbol{Z}}_K \in \mathbb{R}^{N\times 784}. \tag{15}$$

To qualitatively assess reconstruction quality, a few sample handwritten digits are reconstructed using the selected number of principal components. Figure 5 compares the original images with their PCA-based reconstructions.

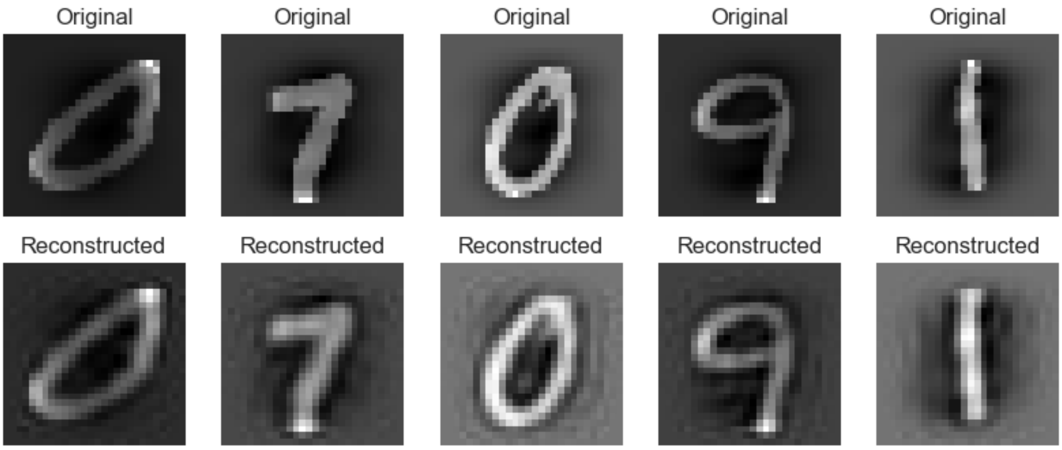

Figure 5: Comparison between original MNIST images and their PCA reconstructions.

It can be observed that the reconstructed images retain the primary structural features of the digits while losing fine-grained details. This confirms that PCA preserves dominant variance directions while discarding high-frequency variations and noise.

6. Residual Computation

After reconstructing with K principal components, the residual matrix is computed as:

$$\boldsymbol{R} = \boldsymbol{Z} - \widehat{\boldsymbol{Z}}_K = \boldsymbol{Z} - \boldsymbol{Z}\boldsymbol{V}_K\boldsymbol{V}_K^\top = \boldsymbol{Z}\left(\boldsymbol{I} - \boldsymbol{V}_K\boldsymbol{V}_K^\top\right). \tag{16}$$

Each residual vector $\boldsymbol{r}_i$ is the part of sample i not captured by the PCA subspace.

6.1 Residual Variance (Q-statistic)

The notebook computes a per-sample residual energy:

$$Q_i = \|\boldsymbol{r}_i\|_2^2 = \sum_{j=1}^{784} r_{ij}^2. \tag{17}$$

Large Q_i indicates that the PCA reconstruction misses substantial information for that sample.

6.2 Residual Energy Analysis

To quantify the information not captured by the PCA subspace, the residual energy Q_i was computed for each sample. This provides a measure of reconstruction loss after dimensionality reduction.

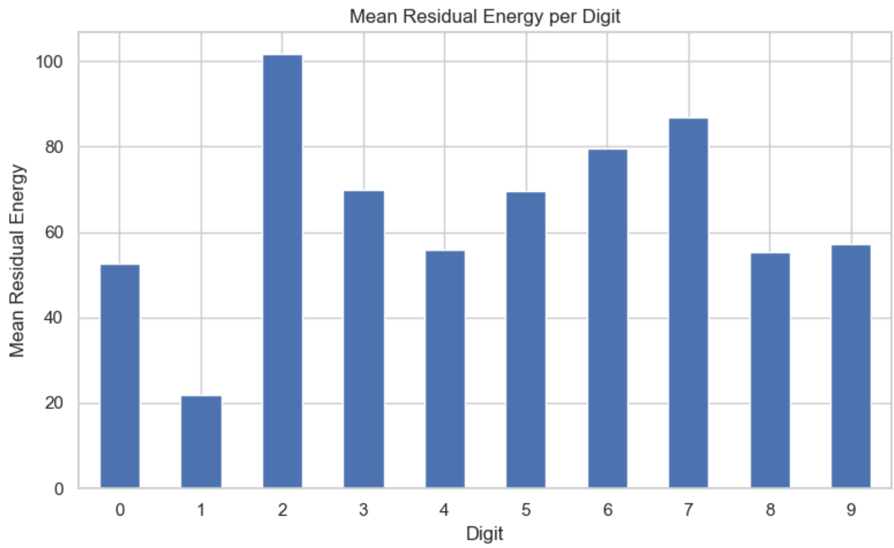

Figure 6: Mean residual energy per digit class.

Figure 6 shows the average residual energy grouped by digit label. Digit "2" exhibits the highest residual energy because it has curved and diagonal complexity. In contrast, digit "1" is structurally simple, mostly vertical stroke. PCA struggles more with shape variability

6.3 Residual PCA (PCA on the Residual Space)

PCA is applied again, but now on the residual matrix $\boldsymbol{R}$.

6.3.1 Residual Covariance Matrix

$$\boldsymbol{C}_R = \frac{1}{N-1}\boldsymbol{R}^\top \boldsymbol{R}. \tag{18}$$

6.3.2 *Eigen-Decomposition*

$$\boldsymbol{C}_R \boldsymbol{u}_m = \gamma_m \boldsymbol{u}_m, \qquad m = 1, 2, \ldots, 784, \tag{19}$$

where γ_m are residual eigenvalues and $\boldsymbol{u}_m$ are residual principal axes.

Choose the top K_2 residual components:

$$\boldsymbol{U}_{K_2} = \begin{bmatrix} \boldsymbol{u}_1 & \cdots & \boldsymbol{u}_{K_2} \end{bmatrix} \in \mathbb{R}^{784 \times K_2}. \tag{20}$$

The residual PCA embedding (scores) is:

$$\boldsymbol{T}_R = \boldsymbol{R}\boldsymbol{U}_{K_2} \in \mathbb{R}^{N \times K_2}. \tag{21}$$

6.4 Residual Scree Plot

To analyze the structure of the residual space, PCA was applied to the residual matrix R. The eigenvalue spectrum of the residual covariance matrix is shown in Figure 7.

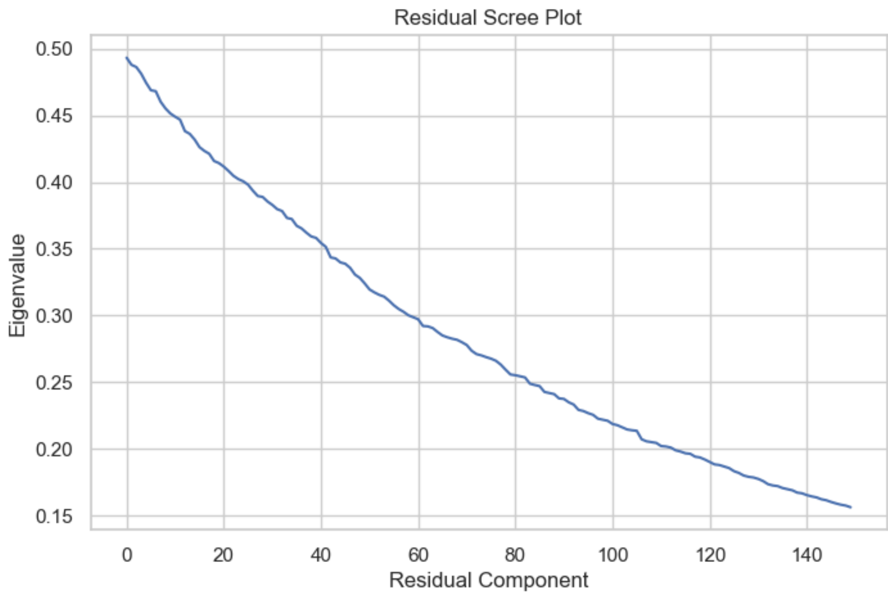

Figure 7: Residual scree plot showing eigenvalues of the residual covariance matrix.

A rapid decay of residual eigenvalues indicates that most structured variance has already been captured by the primary PCA subspace. Non-trivial eigenvalues suggest the presence of secondary structured variability.

6.5 Residual Cumulative Variance

The cumulative explained variance of the residual components is computed as:

$$\mathrm{CEV}_R(K_2) = \sum_{m=1}^{K_2} \frac{\gamma_m}{\sum_j \gamma_j} \tag{22}$$

where γ_m are the eigenvalues of the residual covariance matrix.

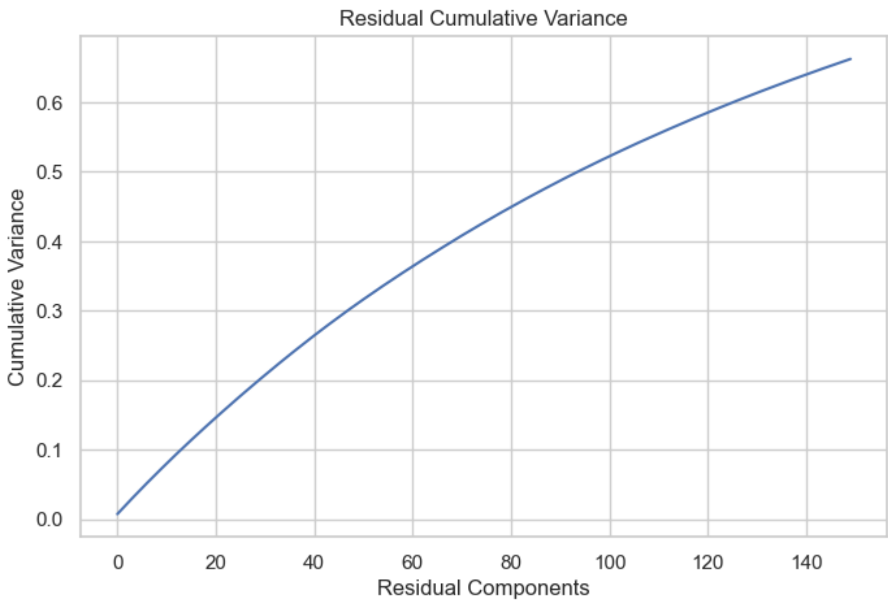

Figure 8: Cumulative explained variance in the residual subspace.

This analysis quantifies how much variability remains in the residual space after primary dimensionality reduction.

7. Observations

The PCA results show that the MNIST dataset cannot be reduced to very few dimensions without losing a lot of information. The first principal component explains only 5.64% of the total variance. To keep 80% of the variance, 149 components are needed, and 330 components are required to keep 95%.

The classification accuracy improves quickly as the number of principal components increases. However, after around $K \approx 250$, the accuracy becomes almost stable. This means that we do not need to keep all the variance to achieve good classification performance.

From the reconstruction results, it is clear that PCA keeps the main shape of the digits but removes small details. The residual energy analysis shows that some digits, such as "2", are more complex and harder to represent compared to simpler digits like "1".

Residual PCA shows that the leftover part still contains some structured information, but it is much smaller compared to the main PCA subspace.

8. Conclusion

In this study, PCA was applied to the MNIST dataset to reduce dimensionality and analyze how much information is preserved after reduction.

The results show that a large number of components are needed to retain most of the total variance. However, classification accuracy becomes stable much earlier, around $K \approx 250$. This means that keeping all variance is not necessary for good prediction performance.

The reconstruction results show that PCA keeps the main shape of the digits but removes small details. The residual analysis shows that some digits (like “2”) are more complex and harder to represent than simpler digits (like “1”).

Overall, PCA is effective for reducing dimensionality while maintaining good classification accuracy.